

Course Introduction

Welcome to Python Programming

Control flow and conditionals

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Conditional statements

This reading introduces you to the conditional statements of **if**, **else** and **elif**.

If

In keeping with the light switch example, the state of the switch can be stored with a Boolean value of **True** or **False**.

- On = True
- Off = False

```
1
2 #Light is currently off
3 current = False
4
5 if current:
6     current = False
7     print('Turning light off')
8
9 if not current:
10    current = True
11    print('Turning light on')
12
13
```

Run
Reset

The code snippet above has a variable called **current** which keeps track of the state of the light - on or off. The first **if** statement will check if the light is on and if it is, the flow will go inside the condition and set the **current** variable to **False** - turning the light off. In the above code snippet, the value of the **current** variable is initially set to **False**, so this condition is not met.

The second **if** statement will check if the light is off and if it is, the flow will go inside the condition and will set the **current** variable to **True** - turning the light on.

If else

The above code works fine but it can be rewritten more effectively by using another condition called **else**. The following code is an example:

```
1 current = False
2
3 if current:
4     current = False
5     print('Turning light off')
6 else:
7     current = True
8     print('Turning light on')
9
```

Run
Reset

The **else** statement has made your code a bit easier to read and given that the flow relates to the same condition, it makes more sense to combine them as part of a single unit.

elif

Python also has another condition called **elif** which helps when you have multiple conditions to satisfy. The light switch example is pretty straightforward in that you only have to check for the state of on or off - **True** or **False**. In certain conditions, it may not be that easy. Thankfully **elif** is here to help.

Let's say you want to give a certain discount to customers if they spend over \$100. You will also provide an extra discount if that customer is part of a loyalty program. If the customer is not part of the loyalty program and did not spend over a \$100, a service charge of 5% is applied.

```
1 loyalty_customer = True
2 total_bill = 124
3
4 if loyalty_customer and total_bill > 100:
5     #give 20% discount
6     total_bill = total_bill - (float(total_bill)/ 100) * 20
7 elif total_bill > 100:
8     #give 10% discount
9     total_bill = total_bill - (float(total_bill)/ 100) * 10
10 else:
11     #sorry no discount, 5% service charge applied.
12     print('Sorry, no discount ...')
13
14 print('Total Bill: ', float(total_bill))
```

Run
Reset

The above code snippet first checks to see if the customer is part of the loyalty program and if they are spending over \$100. If both conditions are met, a discount of 20% is applied to the bill. The **elif** statement will only be executed if the first **if** condition is not met. The **elif** statement will only check if the bill is over \$100 and if it is, it will apply a discount of 10% to the bill.

The final **else** statement is only executed if neither of the other two conditions are met. In this case, a charge of 5% is applied to the bill.

